

SDC Command Mapping Cheat Sheet

A quick reference guide showing exactly which SDC commands belong in which of the 4 MCMM constraints files.

1. func_ss.sdc (Functional Mode, Slow Corner)

Goal: Fix Setup (Max) Violations at High Heat/Low Voltage.

Command Type	Specific Command to Use
Clock Definition	<code>create_clock -period 2.0 [get_ports clk]</code> (Fast Clock)
Operating Condition	<code>set_operating_conditions ss0p95v125c</code> (Slow/Hot Library)
I/O Delays	<code>set_input_delay -max ...</code> <code>set_output_delay -max ...</code> (Setup/Max Timing)
Mode Control	<i>No scan_en constraint needed (chip is in normal operation).</i>

2. func_ff.sdc (Functional Mode, Fast Corner)

Goal: Fix Hold (Min) Violations at Freezing Cold/High Voltage.

Command Type	Specific Command to Use
Clock Definition	<code>create_clock -period 2.0 [get_ports clk]</code> (Fast Clock)
Operating Condition	<code>set_operating_conditions ff0p95vm40c</code> (Fast/Cold Library)
I/O Delays	<code>set_input_delay -min ...</code> <code>set_output_delay -min ...</code> (Hold/Min Timing)
Mode Control	<i>No scan_en constraint needed.</i>

3. scan_ss.sdc (Scan Mode, Slow Corner)

Goal: Ensure the tester can shift data in at max delays without setup failures.

Command Type	Specific Command to Use
Clock Definition	<code>create_clock -period 20.0 [get_ports clk]</code> (Slow Tester Clock)
Operating Condition	<code>set_operating_conditions ss0p95v125c</code> (Slow/Hot Library)
I/O Delays	<code>set_input_delay -max ...</code> <code>set_output_delay -max ...</code> (Tester Max Delays)
Mode Control	<code>set_case_analysis 1 [get_ports scan_en]</code> (Force Test Mode)

4. scan_ff.sdc (Scan Mode, Fast Corner)

Goal: Fix Hold violations on the back-to-back scan chains at fast conditions.

Command Type	Specific Command to Use
Clock Definition	<code>create_clock -period 20.0 [get_ports clk]</code> (Slow Tester Clock)
Operating Condition	<code>set_operating_conditions ff0p95vm40c</code> (Fast/Cold Library)
I/O Delays	<code>set_input_delay -min ...</code> <code>set_output_delay -min ...</code> (Tester Min Delays)
Mode Control	<code>set_case_analysis 1 [get_ports scan_en]</code> (Force Test Mode)

Pro-Tip: Global exceptions like `set_false_path` on asynchronous resets usually belong in ALL 4 files, because an impossible physical path is impossible regardless of the weather or the mode!